

ELECTIVE COURSE-5.3

UCS744: AUTONOMOUS SYSTEMS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To become familiar with the concept of robotics and learn different types of algorithms for process uncertainties, data fusion and localization.				
Syllabus Fundamentals Of Robotics & Automation: Intelligent Robots, Control Systems and Components. Robot Motion Analysis and Control, Robot End Effectors, tactile and vision sensors in robotics, Path planning in joint and task space; Obstacle avoidance and optimal planning; Review of robot control methods; Optimization in robotics; Human-robot interaction; joint and link flexibilities; Walking machines and Exoskeletons; Robot hand and multifingered grasp, manipulation and control; Tendon driven manipulator. Sensors and sensing: Model of Sensors & Process uncertainties, Introduction to estimation, estimation methods & relation between different estimators, State space modeling, LTI Systems & Kalman Filter & Extended Kalman Filter, Other Navigation Filters including Bayesian Filters, Information Filters, Particle Filter etc. Sensors used in Robotics: Accelerometer, Gyro, Compass, Encoder, Laser, Ultrasonic Sensor, Camera, Sonar, InfraRed Sensor, Tactile Sensor etc. Multisensor Data Fusion Fundamentals; INS, GPS Aided Navigation & Data Fusion, Future Trends, Applications, Feedback control of dynamic systems. Computer Vision & Robotics: Image Acquisition, Video frames, Key frames, Background subtraction, Moving object detection, Robot Localization, SLAM.				
Laboratory Work To implement various techniques studied during the course.				
Course Learning Outcomes (CLOs) The students will be able to: 1. understand the different robot components and their control systems. 2. understand uncertainty estimation and data fusion algorithms. 3. implement a human-robot interaction system. 4. implement vision based robot localization and path planning.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Text Books

1. Hands-On Robotics Programming with C++: Leverage Raspberry Pi 3 and C++ libraries to build intelligent robotics applications, Dinesh Tavasalkar, PACKT (2019)
2. Multi-Sensor Data Fusion with MATLAB, Jitendra R. Raol ,CRC Press (2010)

Reference Books

Mobile Robot Localization and Map Building, José A. Castellanos, Juan D. Tardós, Springer (2012)

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessional (Assignments/Projects/Tutorials/Quizzes)	35

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